

Cache-Coherence Traffic on a 6×7 Mesh

Uniform (distributed directory) vs. Hotspot (4 centralized directories)

BookSim2 · anynet 42-node mesh · request/reply · write fraction 0.3 · 1-flit control / 5-flit data · 2 subnets · 4 VCs

0.40

Uniform peak throughput
flits/node/cyc

0.14

Hotspot peak throughput
flits/node/cyc

0.0672

Uniform saturation load
pkt/node/cyc

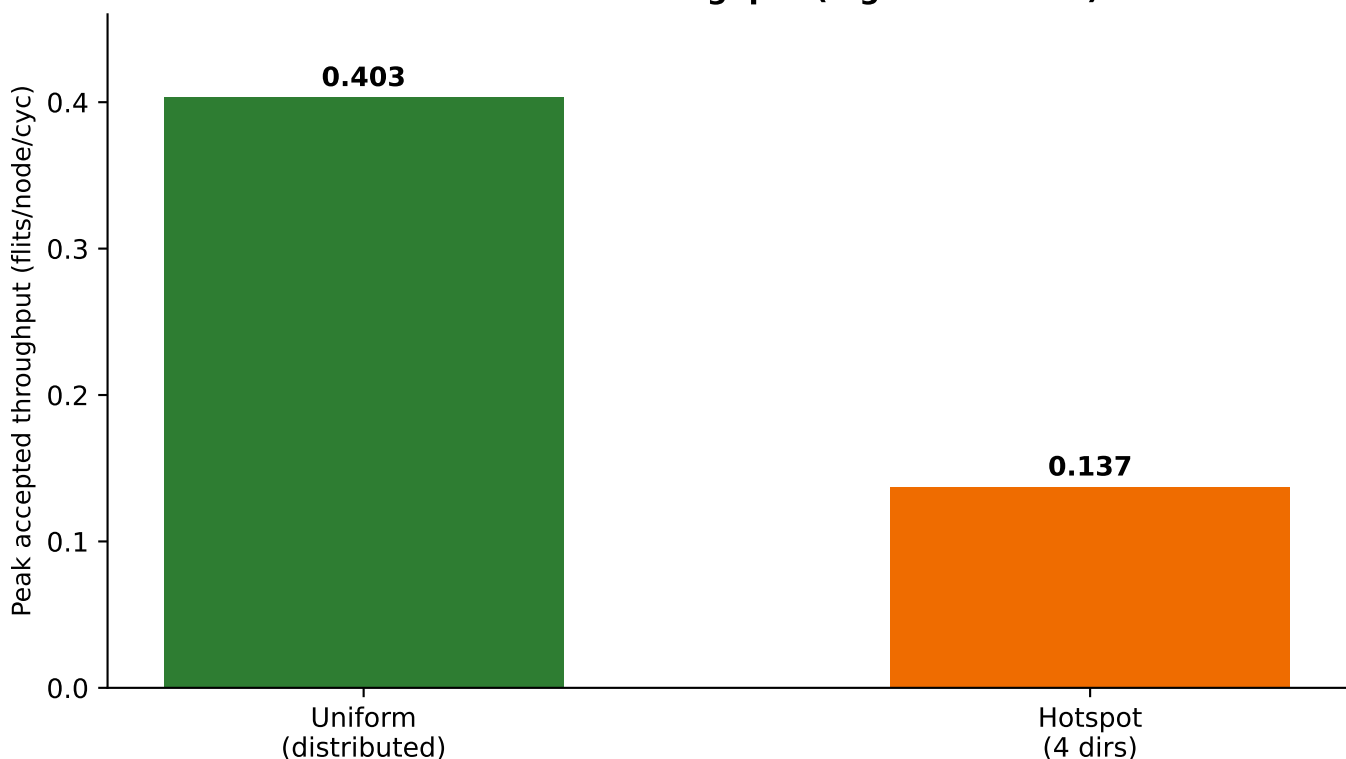
0.0234

Hotspot saturation load
pkt/node/cyc

Key findings

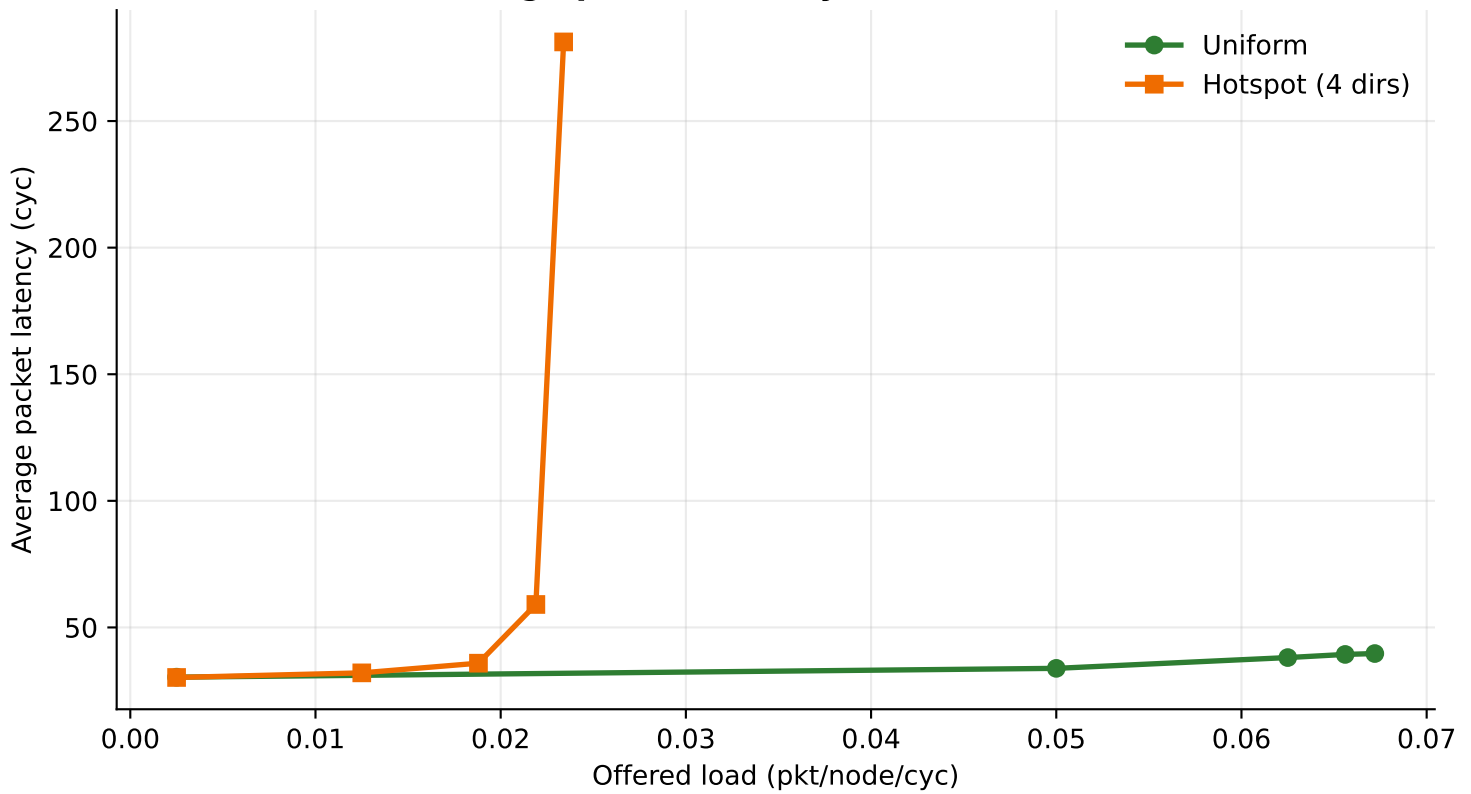
- At low load both patterns behave identically (avg hop count ≈ 5.1 , zero-load latency ≈ 30 cyc).
- Distributing directories across all nodes sustains ~ 0.40 flits/node/cyc; funneling all requests into 4 hotspot banks collapses this to ~ 0.14 — about $2.9\times$ lower.
- Hotspot saturates at $\sim 1/3$ the offered load and its latency blows up sharply (to 281 cyc), because the hotspot nodes' ejection ports and the links feeding them become the bottleneck.
- Conclusion: a centralized coherence directory is a throughput bottleneck; distributing the home nodes is essential for scalable coherence traffic.

Peak sustained throughput (higher is better)

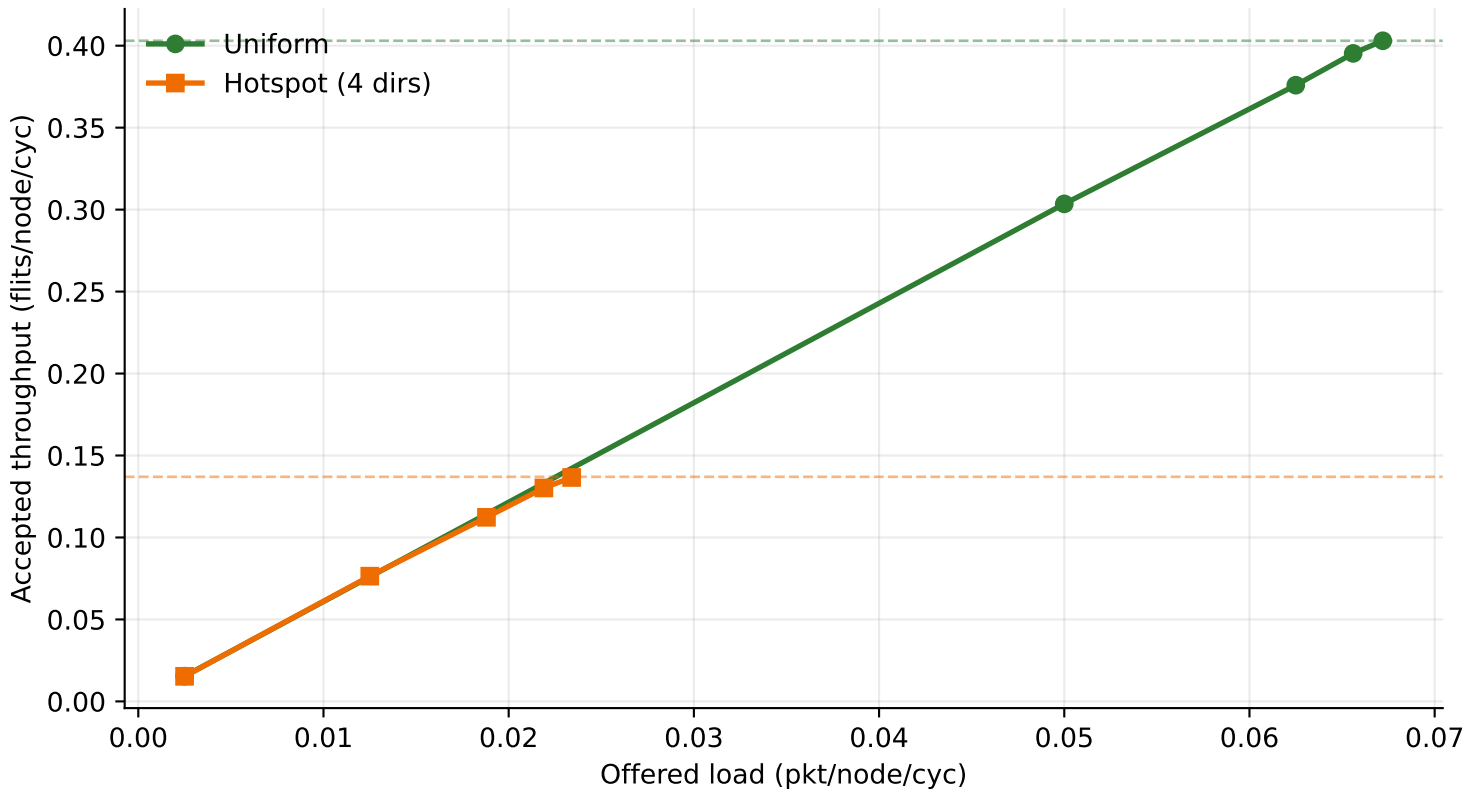


Latency-Throughput Characteristics

Average packet latency vs. offered load



Accepted throughput vs. offered load



Sweep data points

Uniform			Hotspot (4 dirs)		
load	thrpt	lat	load	thrpt	lat
0.0025	0.0154	30.36	0.0025	0.0153	30.24
0.0500	0.3035	33.84	0.0125	0.0764	32.06
0.0625	0.3759	38.10	0.0188	0.1123	35.88
0.0656	0.3953	39.34	0.0219	0.1302	59.07
0.0672	0.4030	39.69	0.0234	0.1366	281.25